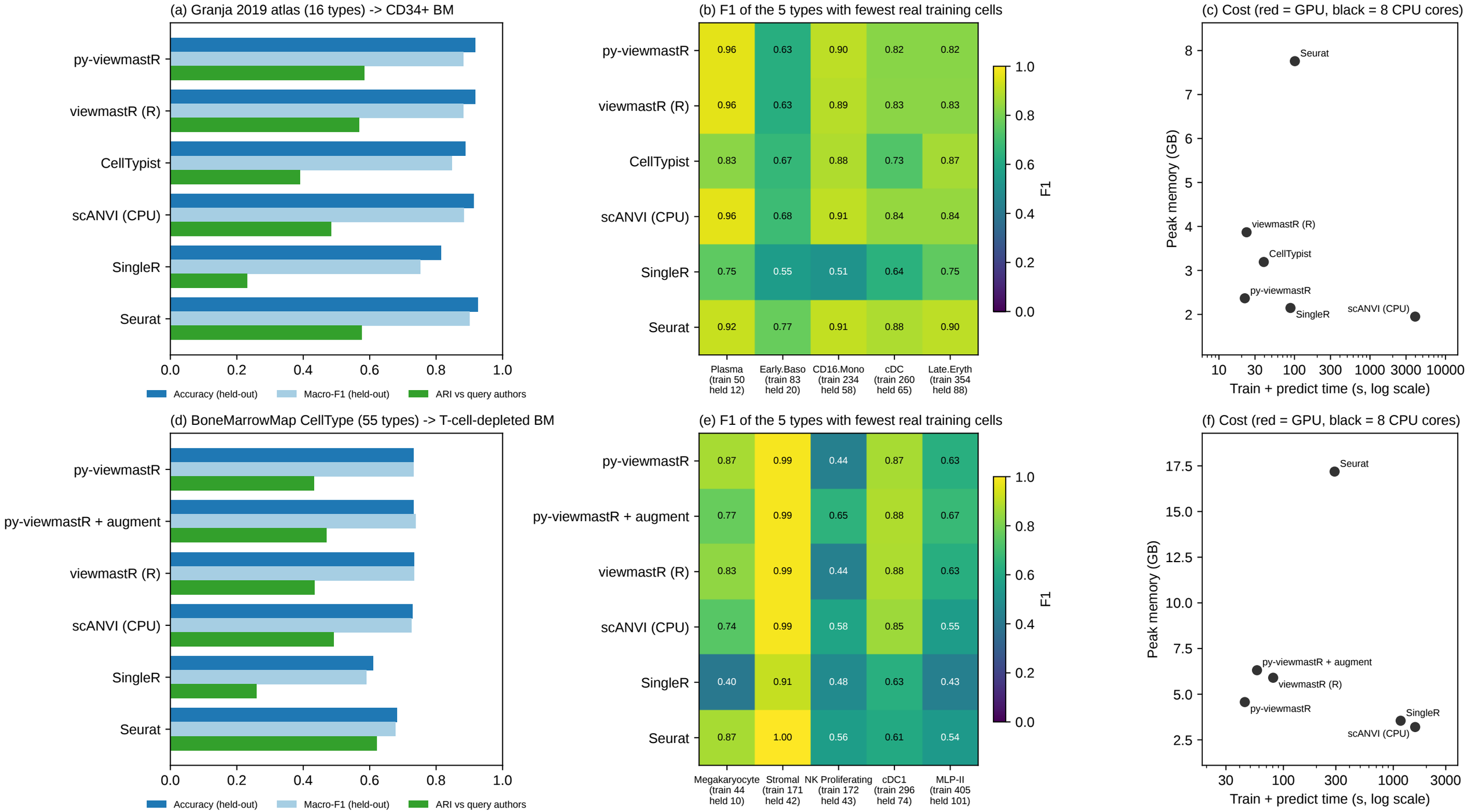
(a) Granja 2019 atlas (16 types) -> CD34+ BM
py-viewmastR
viewmastR (R)
CellTypist
scANVI (CPU)
SingleR
Seurat
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy (held-out)
Macro-F1 (held-out)
ARI vs query authors
(b) F1 of the 5 types with fewest real training cells
0.96 0.63 0.90 0.82 0.82
0.96 0.63 0.89 0.83 0.83
0.83 0.67 0.88 0.73 0.87
0.96 0.68 0.91 0.84 0.84
0.75 0.55 0.51 0.64 0.75
0.92 0.77 0.91 0.88 0.90
Plasma (train 50 held 12)
Early.Baso (train 83 held 20)
CD16.Mono (train 234 held 58)
cDC (train 260 held 65)
Late.Eryth (train 354 held 88)
F1
(c) Cost (red = GPU, black = 8 CPU cores)
Peak memory (GB)
Seurat
viewmastR (R)
CellTypist
py-viewmastR
SingleR
scANVI (CPU)
10
30
100
300
1000
3000
10000
Train + predict time (s, log scale)
(d) BoneMarrowMap CellType (55 types) -> T-cell-depleted BM
py-viewmastR
py-viewmastR + augment
viewmastR (R)
scANVI (CPU)
SingleR
Seurat
(e) F1 of the 5 types with fewest real training cells
0.87 0.99 0.44 0.87 0.63
0.77 0.99 0.65 0.88 0.67
0.83 0.99 0.44 0.88 0.63
0.74 0.99 0.58 0.85 0.55
0.40 0.91 0.48 0.63 0.43
0.87 1.00 0.56 0.61 0.54
Megakaryocyte (train 44 held 10)
Stromal (train 171 held 42)
NK Proliferating (train 172 held 43)
cDC1 (train 296 held 74)
MLP-II (train 405 held 101)
(f) Cost (red = GPU, black = 8 CPU cores)
Peak memory (GB)
Seurat
py-viewmastR + augment
viewmastR (R)
py-viewmastR
SingleR
scANVI (CPU)
30
100
300
1000
3000
Train + predict time (s, log scale)